nature geoscience

Article

https://doi.org/10.1038/s41561-026-02057-3

Central role of Labrador Sea convection for transport of oxygen into the deep North Atlantic Ocean

Received: 31 August 2025

Accepted: 26 June 2026

Published online: 17 August 2026

Check for updates

Una Kim Miller[1,11] ✉, Jaime Palter[1], Ellen Park[2,3], Dariia Atamanchuk[4], Kristen Fogaren[5], Yao Fu[6], Johannes Karstensen[7], Jannes Koelling[8,9], Isabela Le Bras[10], Hiroki Nagao[3,10], David Nicholson[2], Hilary Palevsky[5] & Meg Yoder[5]

Convection in the subpolar North Atlantic feeds highly oxygenated water into the lower limb of the Atlantic Meridional Overturning Circulation (AMOC), sustaining deep-sea organisms across a vast expanse of the ocean. Yet, the rates and processes responsible for this oxygenation have remained largely hidden by a lack of direct measurements. Here we apply newly developed calibration protocols for multi-year oxygen measurements to quantify oxygen transport across a mooring array spanning the Labrador Sea. The results show that Labrador Sea deep convection results in a substantial export of oxygen to the lower limb of the AMOC. Specifically, local air–sea exchange adds ~4.1 Tmol yr^{-1} of oxygen to dense Labrador Sea waters, while deep convection moves ~23.5 Tmol yr^{-1}, added to lighter waters upstream in the eastern subpolar North Atlantic, to those same dense waters. We estimate that this export is sufficient to balance biological respiration across much of the deep North Atlantic, making processes in the Labrador Sea critical to North Atlantic oxygenation despite their small contribution to the densification of the AMOC lower limb. This finding shows that future risk of deep North Atlantic deoxygenation depends not only on the processes that drive overturning, but also on Labrador Sea deep convection.

The availability of oxygen in the ocean shapes the health and geographic distributions of marine organisms[1,2]. Rising global temperatures have driven persistent and widespread oxygen loss in the ocean[3,4], yet the North Atlantic has seen little change in its oxygen inventory since observations began in the 1960s[5,6]. This stability has been attributed to the continued injection of oxygenated waters to the deep ocean by the Atlantic Meridional Overturning Circulation (AMOC) (for example, ref. 7), a current system broadly defined by the transformation of a warm, northward-flowing upper limb of ocean currents into a cold, southward-flowing lower limb. In the upper limb, the AMOC transports surface waters through the Gulf Stream and North Atlantic Current (Fig. 1a). These waters enter a series of boundary currents that encircle the cyclonic North Atlantic subpolar gyre, with a portion entering the Nordic Seas to the north. Along this route, ocean heat loss to the atmosphere and freshwater input from the Arctic transform the surface waters into colder, fresher and denser water masses[8–10], which then return equatorward at depth via the Deep Western Boundary Current and eddy-driven pathways in the basin interior[11,12] (Fig. 1a).

Building on prior work in the region, recent observations from the Overturning in the Subpolar North Atlantic Program (OSNAP[13]; Fig. 1a) mooring array showed that both the mean and seasonal-to-interannual variability of the subpolar AMOC is largely determined by processes

A full list of affiliations appears at the end of the paper. ✉e-mail: umiller@cornell.edu

https://doi.org/10.1038/s41561-026-02057-3

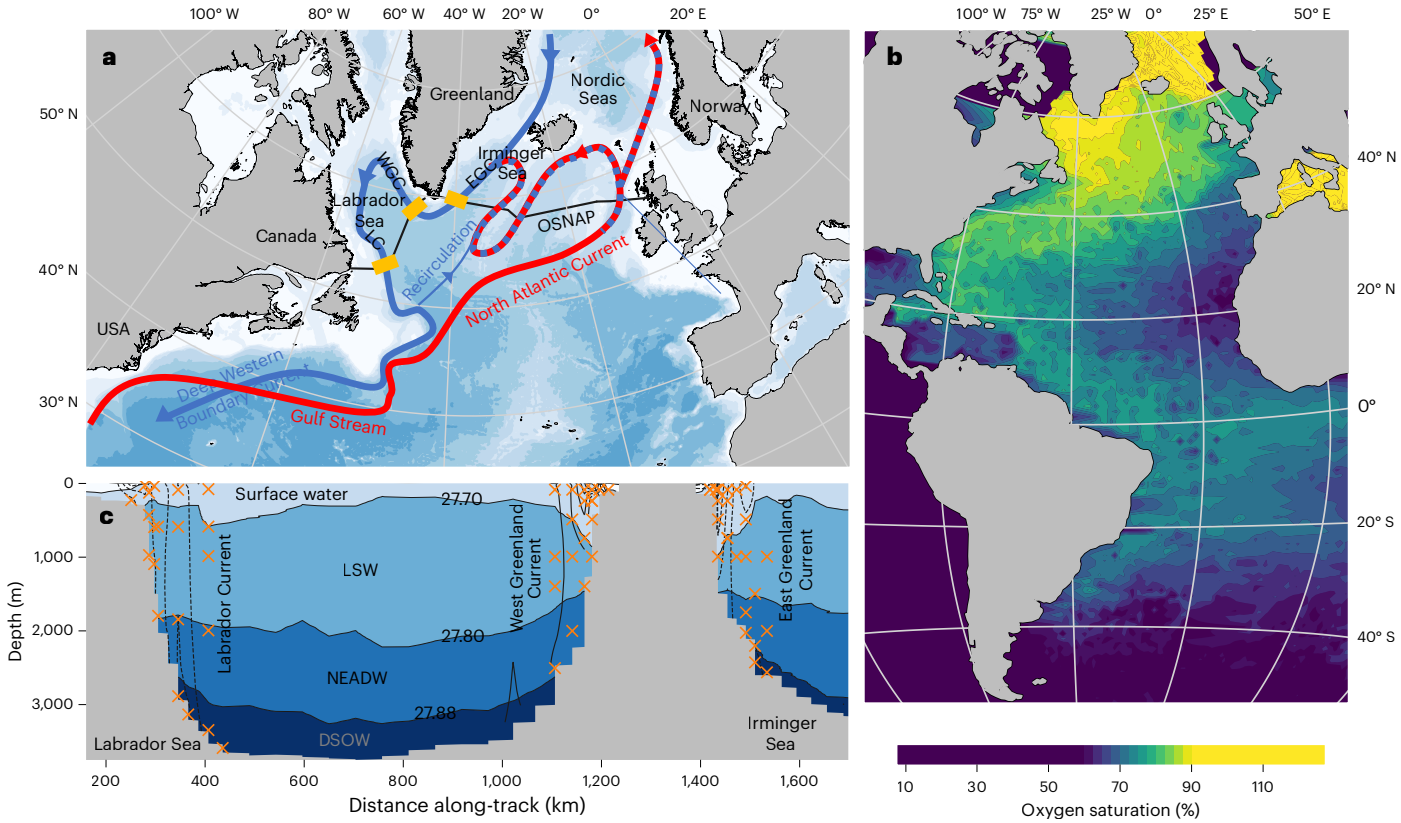


Fig. 1 | Deployment setting of oxygen optodes to quantify the contribution of the Labrador Sea to the ventilation of the lower limb of the AMOC. **a**, Circulation in the subpolar North Atlantic with the OSNAP line shown in black. The thick yellow sections indicate where oxygen optodes were deployed over 2020–2022 in and near the boundary currents entering (East Greenland Current and West Greenland Current, EGC and WGC) and exiting (Labrador Current, LC) the Labrador Sea. Red (blue) ocean current pathways indicate warm, shallow (cold, deep) currents. Dashed red/blue segments indicate progressive densification. **b**, Depth-averaged oxygen saturation (%) calculated within the LSW potential density class ($27.7 < \sigma_\theta < 27.8$ kg m^{-3}) from 1965–2022 World Ocean Atlas climatology[51,52]. **c**, A depth cross-section view of optode deployments along the OSNAP line. Optodes are marked with 'x's and dashed (solid) contours indicate currents from the OSNAP velocity grid product flowing out of (into) the page at intervals of 0.1 m s^{-1}. Filled isopycnal contours outline surface waters ($\sigma_\theta < 27.7$ kg m^{-3}), LSW, Northeast Atlantic Deep Water (NEADW; $27.8 < \sigma_\theta < 27.88$ kg m^{-3}) and DSOW ($\sigma_\theta > 27.88$ kg m^{-3}). Basemap data in **a** from Natural Earth (https://www.naturalearthdata.com).

in the eastern subpolar gyre and Nordic Seas, with little contribution from the Labrador Sea[14]. Time series from the first 6 years of OSNAP (2014–2020) show that overturning east of Greenland is five times greater than that in the Labrador Sea[13,15,16] and explains upwards of 80% of the monthly variance in the strength of the AMOC[17].

The Labrador Sea is a site of intensive, open-ocean deep convection[18,19], yet the density-compensating effects of freshening and cooling minimize its contributions to the AMOC[20]. Here, frigid winds extract heat from the surface ocean. But simultaneous freshening, due to discharge of Arctic and Greenland meltwater and an excess of precipitation over evaporation[21], nearly counteracts the influence of cooling on density. The result is a small net density increase that can support deep mixing in a weakly stratified water column but represents a minimal overturning transport in density space of only ~2 Sv ($1\ \text{Sv} = 10^6\ \text{m}^3\ \text{s}^{-1}$)[15].

Despite its relatively minimal influence on AMOC strength and variability, there is reason to believe that the Labrador Sea could play an important role in deep North Atlantic ventilation[22–27]. Ventilation is the injection of surface waters rich in oxygen and other atmospheric gases into the ocean interior[28], and is central to projecting the vulnerability of the deep North Atlantic to large-scale deoxygenation. The thickness of mixed layers attained by Labrador Sea deep convection can exceed 1,500 m, exposing a large volume of low-oxygen deep water directly to the atmosphere. The resulting air–sea gradient in oxygen concentration drives a flux of oxygen into the ocean that is further enhanced by vigorous, bubble-mediated air–sea gas exchange[29] and an increase in gas solubility resulting from the cooling of surface waters. Additionally, oxygen accumulated in the surface ocean throughout the eastern subpolar North Atlantic finally achieves the density that separates the upper from lower limb of the AMOC through deep convection in the Labrador Sea (and, to some extent, in the Irminger Sea)[30]. Accordingly, Labrador Sea deep convection produces a water mass known as Labrador Sea Water (LSW) that is traceable at intermediate depths along the Deep Western Boundary Current by not only its immense thickness (and hence low potential vorticity signal[31]), but also by its high oxygen saturation (Fig. 1b)[32]. Here we show results from a new set of high-quality, full-water column oxygen time series data across both boundary currents entering and exiting the Labrador Sea (Fig. 1a,c), enabling us to robustly quantify the contribution of the Labrador Sea to North Atlantic ventilation.

Observing oxygen transport

In June 2020, the Gases in the Overturning and Horizontal circulation of the Subpolar North Atlantic Program (GOHSNAP) deployed 60 oxygen optodes on the segments of the OSNAP mooring array in the boundary currents of the Labrador and western Irminger Seas (Fig. 1c)[33]. The 24-month time series data were recovered in July 2022 and calibrated against shipboard dissolved oxygen profiles collected upon both deployment and recovery of the moorings, following the protocols we developed for the calibration of moored optode data (Methods)[34]. The discrete oxygen measurements were gridded along the OSNAP line to match the resolution of the OSNAP velocity product[35] using a

https://doi.org/10.1038/s41561-026-02057-3

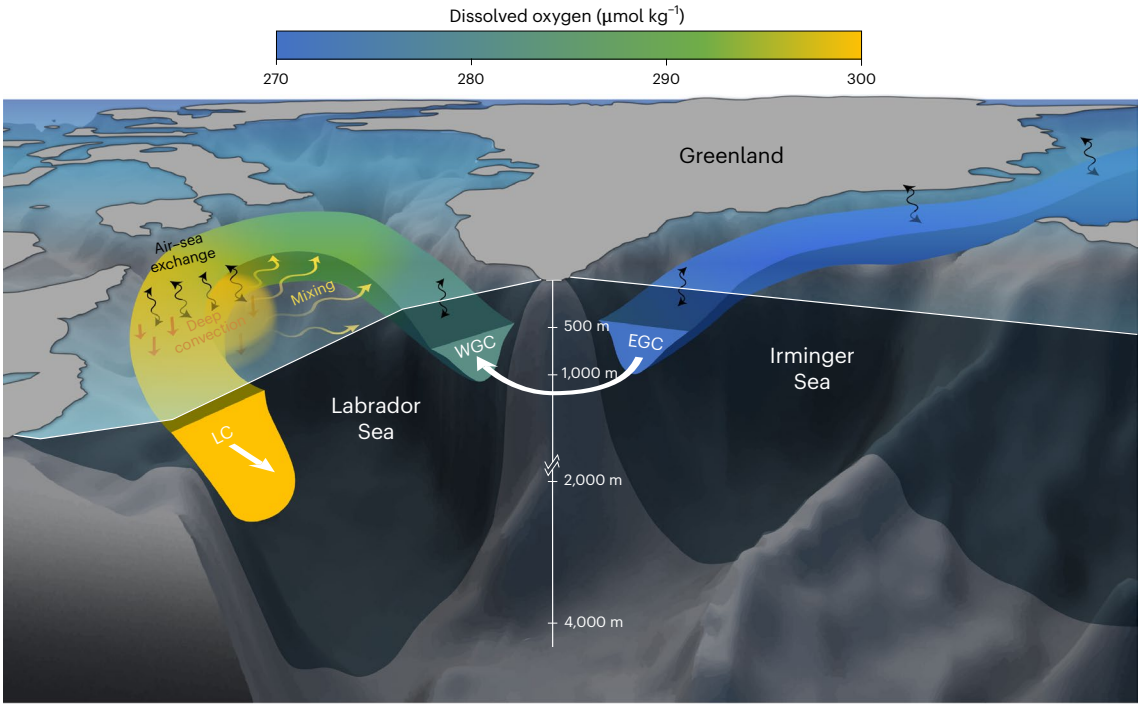


Fig. 2 | Schematic of transport at the density of LSW through the East Greenland Current, West Greenland Current and Labrador Current. Vigorous air–sea gas exchange and deep convection in the Labrador Sea thickens and oxygenates the LSW ($27.7 < \sigma_\theta < 27.8$ kg m^{-3}) layer as it passes through the Labrador and West Greenland boundary currents, thus contributing to the ventilation of the lower limb of the AMOC. Not depicted here is the intermittent deep convection in the Irminger Sea that can result in oxygenation of the LSW density class upstream of the Labrador Sea. Credit: Natalie Renier.

random forest trained on the mooring oxygen with temperature, salinity, latitude, longitude, bathymetry and time as predictor variables, and with uncertainties rigorously quantified (Methods).

Oxygenation of LSW despite minimal changes in density

Waters in the LSW density class ($27.7 < \sigma_\theta < 27.8$ kg m^{-3}; ref. 20; where σ_θ is the potential density anomaly referenced to the sea surface) enter the Labrador Sea through the West Greenland Current and exit as a thicker and more oxygenated layer in the Labrador Current (Fig. 2): LSW within the Labrador Current, downstream of the region where deep convection occurs, is cooler, fresher and on average 15 µmol kg^{-1} higher in oxygen than waters of the same density class in the East and West Greenland Currents (Fig. 3a and Supplementary Fig. 2), upstream of the site of deep convection. This substantial addition of oxygen at depth despite minimal changes in density is a result of the density-compensated nature of deep convection in the Labrador Sea, in which simultaneous cooling and freshening result in little net change in density. Water in the LSW density class flowing through the West Greenland Current is also slightly cooler, fresher and more oxygenated than that in the East Greenland Current (Fig. 3a), probably reflecting input of LSW into the West Greenland Current from the Labrador Sea interior, as depicted by the 'mixing' arrows in Fig. 2.

Addition of oxygen by deep convection

The net 24-month average oxygen transport across the Labrador Sea OSNAP line is −4.1 ± 1.6 Tmol yr^{-1} (mean ± standard deviation, calculated using a Monte Carlo approach, see Methods for details; Fig. 4), meaning the Labrador Sea takes up roughly 4.1 Tmol yr^{-1} of oxygen from the atmosphere and exports it in waters flowing southward. Because we remove the Davis Strait throughflow from our calculations (Methods), the −4.1 Tmol yr^{-1} is not the true net total southward transport of oxygen. However, as the focus of this paper is the ventilation of LSW by deep convection, with which the largely buoyant, shelf-bound Davis

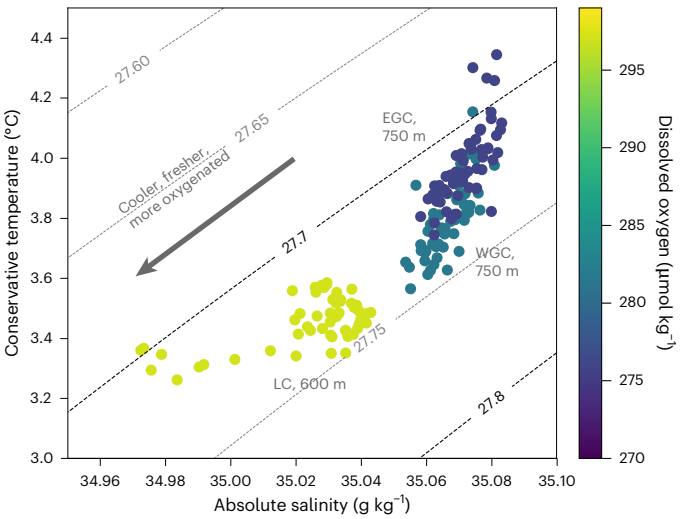


Fig. 3 | Labrador Sea deep convection cools, freshens and oxygenates the LSW layer without substantial changes in density. Temperature–salinity diagram of weekly average measurements from individual optodes at 600 m in the Labrador Current (yellowish-green), 750 m in the West Greenland Current (turquoise) and 750 m in the East Greenland Current (blue) over the 2020–2022 deployment period. Data points are coloured by the 2-year average oxygen concentration measured by each optode. Potential density contours are shown in 0.05 kg m^{-3} intervals in grey, with the upper and lower density thresholds of LSW shown in black.

Strait transport does not interact[36], we continue to use the term 'net' throughout this paper while asking the reader to keep in mind the exclusion of the throughflow component.

To explore the role of deep convection in this export, we partition the oxygen transport calculation into its overturning and isopycnal

https://doi.org/10.1038/s41561-026-02057-3

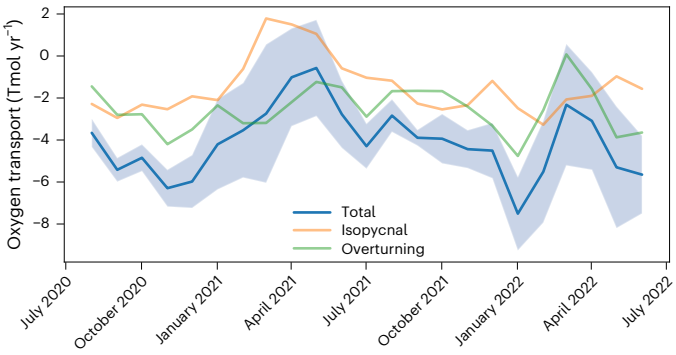


Fig. 4 | Net monthly oxygen transport in the Labrador Sea. Net transport is shown in blue, calculated from the monthly gridded GOHSNAP dissolved oxygen and OSNAP velocity (equation (2); Methods; see Code availability). Shading denotes one standard deviation of transport estimates obtained from Monte Carlo simulations (Methods). The isopycnal (equation (6)) and overturning (equation (5)) components of the total net transport are shown in orange and green, respectively. Negative values indicate export out of the Labrador Sea.

components (Methods). Overturning represents net changes in oxygen transport accompanied by net changes in density, while the isopycnal term represents net changes in oxygen transport along surfaces of constant density. We find that −1.5 ± 0.7 Tmol yr^{-1}, or ~40%, of the −4.1 Tmol yr^{-1} time-averaged net oxygen transport occurs through isopycnal oxygenation. Notably, this partitioning means that a large portion of the net export of oxygen occurs independently of density changes, and represents an instance where ventilation is partially decoupled from overturning[28]. A cumulative-sum plot of time-averaged isopycnal transport from light to heavy density classes ($27.7 < \sigma_\theta < 27.8$ kg m^{-3}) reveals that virtually all of this isopycnal transport occurs in the LSW layer, confirming its association with LSW formation through density-compensated deep convection (Fig. 5a). The remaining −2.6 ± 2.0 Tmol yr^{-1} (Fig. 5b), or ~60%, of the net oxygen transport occurs through overturning, which, in the Labrador Sea, is dominated by the conversion of surface waters to LSW[20].

We note that the approximate 40/60 partitioning of isopycnal and overturning transport is sensitive to uncertainties in the monthly oxygen and velocity fields (Methods); Monte Carlo iterations yield an interquartile range of 21/79% to 63/37%. Furthermore, the relative importance of isopycnal and overturning transport varies month to month (Fig. 4), suggesting a 2-year average may not hold over longer timescales. Still, our results suggest a pathway for oxygen to the AMOC lower limb that is independent of overturning.

Redistribution of oxygen transport

In addition to taking up a net 4.1 ± 1.6 Tmol yr^{-1} of oxygen locally and injecting it into the lower AMOC limb by way of LSW formation, Labrador Sea mixing (including that driven by deep convection) plays a second, important role in North Atlantic ventilation. The cumulative-sum plot of overturning transport shows a change of 23.5 ± 7.3 Tmol yr^{-1} within the LSW density class (Fig. 5b). This oxygen transport gain within LSW illustrates that 23.5 Tmol yr^{-1} of oxygen enter the Labrador Sea within buoyant surface layers and leave within the LSW layer due to the density transformation associated with ~2 Sv of overturning (Supplementary Fig. 1). The small overturning cell at $\sigma_\theta > 27.83$ kg m^{-3} probably represents a transformation of Northeast Atlantic Deep Water into Denmark Strait Overflow Water (DSOW)[20], which results in a small redistribution of oxygen from the former to the latter layer. Unlike the net transport of −4.1 Tmol yr^{-1}, which integrates the entire water column, overturning transport associated with the LSW layer does not represent an addition of oxygen to the water column, but rather a redistribution of the oxygen transport from the upper limb of the AMOC to the LSW layer, which is part of the lower limb.

Recent work has suggested that Labrador Sea deep convection performs the crucial function of injecting oxygen accumulated in the upper AMOC limb down into the lower limb[30]. Much as the surface waters that constitute the upper limb of AMOC become progressively denser as they travel along the pathways of the subpolar gyre (for example, ref. 37), they also become progressively more oxygenated through air–sea interaction (Fig. 2). Ref. 30 estimated this addition as 24.3 ± 6.9 Tmol yr^{-1} across the eastern subpolar gyre. Yet, much of this oxygen is only able to enter the lower limb through deep convection in the Labrador Sea. Our analysis allows us to quantify this injection rate as 23.5 ± 7.3 Tmol yr^{-1}, suggesting that the redistribution of the oxygen transport from light layers to the LSW layer is a key mechanism for injecting the oxygen added in the east to the deep limb of the AMOC.

Ecological importance of LSW ventilation

Our findings underpin the dual importance of the Labrador Sea in oxygenating the lower limb of AMOC: the Labrador Sea takes up approximately 4.1 ± 1.6 Tmol yr^{-1} of oxygen locally through air–sea exchange and adds it directly to the LSW layer. In addition, approximately 2 Sv of Labrador Sea overturning act to redistribute 23.5 ± 7.3 Tmol yr^{-1} of oxygen accumulated in surface waters along the pathway of the subpolar gyre into the LSW layer. Altogether, a total of 27.6 Tmol yr^{-1} are exported from the basin in the LSW layer.

How crucial is the supply of oxygen from the Labrador Sea in sustaining deep-sea ecosystems? Much of the LSW exported from the Labrador Sea travels through the Deep Western Boundary Current (here referred to as the flow of intermediate and deep waters denser than $\sigma_\theta = 27.7$ kg m^{-3}) into the North Atlantic, with a portion recirculating back into the eastern subpolar gyre (Fig. 1a)[38]. Along both circulation pathways, oxygen within the LSW layer is consumed via biological respiration and lost through mixing with neighbouring water masses. In order to provide an order-of-magnitude estimate of the fate of oxygen added to the LSW layer, we use publicly available respiration rate outputs from the IMPLicit Ecosystem and organic matter Transport and RemIneralization Model (SIMPLE-TRIM) model, which assimilates satellite and oceanographic tracer observations to constrain rates of production and respiration[39]. Roughly 3.7 Tmol yr^{-1} are consumed across the entire North Atlantic from 50° N down to the Equator within the 3–4 °C temperature class of LSW. Note that we use temperature to define the LSW layer, as salinity and density are not available model outputs. This value is comparable to respiration estimates of 2.2–3.8 Tmol yr^{-1} in ref. 22, derived using both fixed and depth-dependent respiration rates summed across an 800-m-thick LSW layer spanning 50° N to the Equator. An additional 6.1 Tmol yr^{-1} are consumed in the eastern subpolar North Atlantic, excluding the Labrador Sea (Methods and Supplementary Fig. 4).

To account for the loss of oxygen from the exported LSW layer due to mixing, we can assume that oxygen mixed into adjacent water masses is eventually consumed through respiration. By widening the range over which respiration is summed from 3–4 °C to 3–6 °C, we include respiration in the overlying low-oxygen thermocline waters. This temperature range does not include the colder underlying Nordic Seas overflow waters that compose the remainder of the AMOC lower limb; this is because overflow waters are similarly high in oxygen to LSW (Supplementary Fig. 3) and thus mixing would not represent a big loss in oxygen for the LSW layer. We note that the formation of the Nordic Seas overflow waters likewise results in an addition of oxygen to the deep limb of the AMOC[40–42]. Summing respiration across both the LSW layer and the thermocline waters in both the eastern subpolar gyre (12.1 Tmol yr^{-1}) and North Atlantic (10.3 Tmol yr^{-1}) yields a sum of ~22 Tmol yr^{-1} (Supplementary Fig. 5). This sum serves as a loose estimate of the downstream oxygen loss term for LSW produced in the Labrador Sea, and is on par with our total calculated export of oxygen within the LSW layer from the Labrador Sea, 27.6 Tmol yr^{-1}.

https://doi.org/10.1038/s41561-026-02057-3

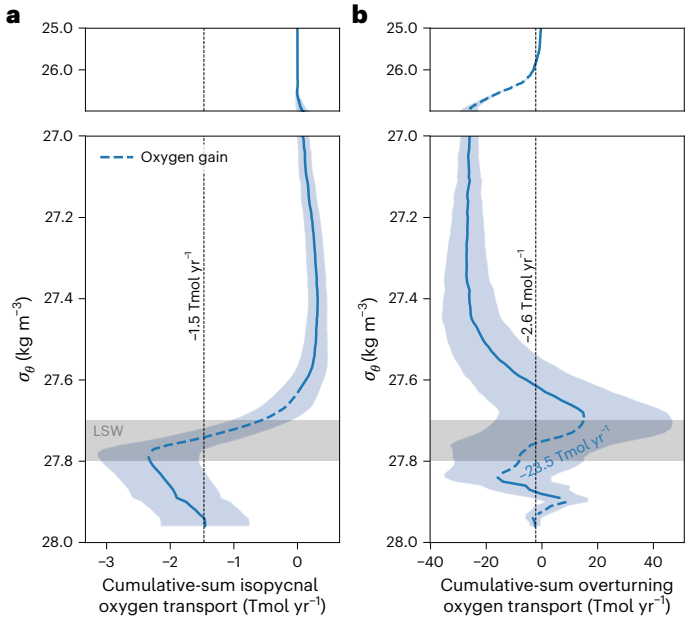


Fig. 5 | Time-averaged isopycnal and overturning oxygen transport cumulatively summed from light to heavy densities. a,**b**, Isopycnal (**a**) and overturning (**b**) oxygen transport. Note the difference in transport range shown on the *x* axis for each panel. The density range for LSW (27.7–27.8 kg m^{-3}) is shaded in grey. Net exports correspond to the final value of the cumulative sums and are marked by vertical dashed lines in each panel. The rate of oxygen gained in the LSW layer through overturning is labelled in **b**. Export of oxygen from the Labrador Sea appears as negative contributions to the cumulative sum and is denoted by dashed lines. Shading denotes one standard deviation of transport estimates obtained from Monte Carlo simulations (Methods).

Our calculations suggest the oxygen exported by the Labrador Sea plays an important role in sustaining downstream biological activity. Notably, this importance to ventilation is despite the Labrador Sea's minor (<10%) contribution to the strength of AMOC[15]. Our findings open the possibility that changes in the supply of oxygen to the deep ocean by the Labrador Sea can have a critical impact on organisms, especially those living at the edge of their metabolic oxygen demand.

Implications for the future of North Atlantic ventilation

The fact that the Labrador Sea injects a substantial amount of oxygen in the AMOC lower limb despite its minor contribution to the strength of AMOC is especially pertinent given the high uncertainty in present and future trends in AMOC strength: some statistical and proxy studies suggest an ongoing weakening of the AMOC driven by warming ocean temperatures and meltwater from the Greenland Ice Sheet, warning of the possibility of a collapse of overturning within this century (for example, refs. 43,44). Yet, observational evidence shows that the strength of the AMOC has been relatively stable for 40–90 years[14,45], and recent numerical modelling studies suggest that collapse before 2100 is unlikely[46,47]. Whatever the fate of AMOC, our results suggest that the full picture of the ventilation of gases in the North Atlantic cannot be understood from the strength of overturning transports alone; variability in Labrador Sea convection is a key component in the oxygenation of the lower limb of AMOC. To illustrate this point, we consider two possible future scenarios for AMOC and Labrador Sea convection strength.

We first explore a scenario in which Labrador Sea deep convection weakens, but AMOC remains largely stable, with a plausible mechanism provided by the increasing freshwater input from the Arctic and Greenland continuing to augment the periodic release of freshwater from the Beaufort Gyre into the Labrador Sea[48,49]. Weakened deep convection in the Labrador Sea would diminish the injection of oxygen into the lower limb of AMOC. Additionally, net oxygen uptake would be reduced. In this scenario, more of the oxygen added along the progressively densifying subpolar gyre surface circulation pathway would be likely to recirculate within the gyre, with limited equatorward connectivity.

In the second scenario, we consider the consequences if Labrador Sea deep convection were to remain stable or strengthen while AMOC weakens. This situation emerged in climate simulations[50] in which freshwater was added in the Nordic Seas to explore its impact on LSW formation. The imposed freshwater flux reduced the simulated formation of the densest water masses flowing out of the Nordic Seas and reduced stratification in the Labrador Sea, thereby allowing for enhanced deep convection. Although the reduced formation of the densest water masses in the Nordic Seas would slow the AMOC, the maintenance of Labrador Sea deep convection would continue adding oxygen to the LSW layer. In this scenario, AMOC slowdown would probably reduce the equatorward export of the deepest water masses, but the supply of oxygen on the LSW layer could remain stable or even increase.

These scenarios highlight the potential for the Labrador Sea to alter the future of oxygen dynamics of the North Atlantic. A better understanding of the processes that control ventilation is especially pressing amid ongoing ocean deoxygenation and its potential to change biogeochemical cycles and disrupt ecosystems. Our findings shift our understanding of how freshwater fluxes in the subpolar North Atlantic may impact the supply of oxygen to the deep ocean, and underscore the utility of long-term biogeochemical and transport observations across and downstream of the subpolar North Atlantic.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41561-026-02057-3.

References

1. Morée, A. L., Clarke, T. M., Cheung, W. W. L. & Frölicher, T. L. Impact of deoxygenation and warming on global marine species in the 21st century. *Biogeosciences* **20**, 2425–2454 (2023).
2. Deutsch, C., Penn, J. L. & Lucey, N. Climate, oxygen, and the future of marine biodiversity. *Annu. Rev. Mar. Sci.* **16**, 217–245 (2024).
3. Oschlies, A., Brandt, P., Stramma, L. & Schmidtko, S. Drivers and mechanisms of ocean deoxygenation. *Nat. Geosci.* **11**, 467–473 (2018).
4. Kwiatkowski, L. et al. Twenty-first century ocean warming, acidification, deoxygenation, and upper-ocean nutrient and primary production decline from CMIP6 model projections. *Biogeosciences* **17**, 3439–3470 (2020).
5. Schmidtko, S., Stramma, L. & Visbeck, M. Decline in global oceanic oxygen content during the past five decades. *Nature* **542**, 335–339 (2017).
6. Ito, T., Minobe, S., Long, M. C. & Deutsch, C. Upper ocean O_2 trends: 1958–2015. *Geophys. Res. Lett.* **44**, 4214–4223 (2017).
7. Tagklis, F., Bracco, A. & Ito, T. Physically driven patchy O_2 changes in the North Atlantic Ocean simulated by the CMIP5 Earth system models. *Glob. Biogeochem. Cycles* **31**, 1218–1235 (2017).
8. Schmidt, S. & Send, U. Origin and composition of seasonal Labrador Sea freshwater. *J. Phys. Oceanogr.* **37**, 1445–1454 (2007).
9. Sicre, M. A. et al. Labrador Current variability over the last 2000 years. *Earth Planet. Sci. Lett.* **400**, 26–32 (2014).
10. Buckley, M. W., Lozier, M. S., Desbruyéres, D. & Evans, D. G. Buoyancy forcing and the subpolar Atlantic Meridional Overturning Circulation. *Phil. Trans. R. Soc. A* **381**, 20220181 (2023).

11. Bower, A. S., Lozier, M. S., Gary, S. F. & Böning, C. W. Interior pathways of the North Atlantic Meridional Overturning Circulation. *Nature* **459**, 243–247 (2009).
12. Koelling, J., Send, U. & Lankhorst, M. Decadal strengthening of interior flow of North Atlantic Deep Water observed by GRACE satellites. *J. Geophys. Res. Oceans* **125**, e2020JC016217 (2020).
13. Lozier, M. S. et al. A sea change in our view of overturning in the subpolar North Atlantic. *Science* **363**, 516–521 (2019).
14. Fu, Y., Li, F., Karstensen, J. & Wang, C. A stable Atlantic Meridional Overturning Circulation in a changing North Atlantic Ocean since the 1990s. *Sci. Adv.* **6**, eabc7836 (2020).
15. Lozier, M. S. Overturning in the subpolar North Atlantic: a review. *Phil. Trans. R. Soc. A* **381**, 20220191 (2023).
16. Fu, Y. et al. Seasonality of the Meridional Overturning Circulation in the subpolar North Atlantic. *Commun. Earth Environ.* **4**, 181 (2023).
17. Li, F. et al. Subpolar North Atlantic western boundary density anomalies and the Meridional Overturning Circulation. *Nat. Commun.* **12**, 3002 (2021).
18. The Lab Sea Group The Labrador Sea Deep Convection Experiment. *Bull. Am. Meteorol. Soc.* **79**, 2033–2058 (1998).
19. Lavender, K. L., Davis, R. E. & Owens, W. B. Observations of open-ocean deep convection in the Labrador Sea from subsurface floats. *J. Phys. Oceanogr.* **32**, 511–526 (2002).
20. Zou, S., Lozier, M. S., Li, F., Abernathey, R. & Jackson, L. Density-compensated overturning in the Labrador Sea. *Nat. Geosci.* **13**, 121–126 (2020).
21. Yang, Q. et al. Recent increases in Arctic freshwater flux affects Labrador Sea convection and Atlantic overturning circulation. *Nat. Commun.* **7**, 10525 (2016).
22. Koelling, J., Atamanchuk, D., Karstensen, J., Handmann, P. & Wallace, D. W. R. Oxygen export to the deep ocean following Labrador Sea Water formation. *Biogeosciences* **19**, 437–454 (2022).
23. MacGilchrist, G. A. et al. Locations and mechanisms of ocean ventilation in the high-latitude North Atlantic in an eddy-permitting ocean model. *J. Climate* **33**, 10113–10131 (2020).
24. Rhein, M., Steinfeldt, R., Kieke, D., Stendardo, I. & Yashayaev, I. Ventilation variability of Labrador Sea Water and its impact on oxygen and anthropogenic carbon: a review. *Phil. Trans. R. Soc. A* **375**, 20160321 (2017).
25. Pickart, R. S., Spall, M. A. & Lazier, J. R. N. Mid-depth ventilation in the western boundary current system of the sub-polar gyre. *Deep Sea Res. I* **44**, 1025–1054 (1997).
26. Brandt, P., Funk, A., Czeschel, L., Eden, C. & Böning, C. W. Ventilation and transformation of Labrador Sea Water and its rapid export in the Deep Labrador Current. *J. Phys. Oceanogr.* **37**, 946–961 (2007).
27. Le Bras, I. A.-A. Labrador Sea Water spreading and the Atlantic Meridional Overturning Circulation. *Phil. Trans. R. Soc. A* **381**, 20220189 (2023).
28. Naveira Garabato, A. C. et al. High-latitude ocean ventilation and its role in Earth's climate transitions. *Phil. Trans. R. Soc. A* **375**, 20160324 (2017).
29. Atamanchuk, D., Koelling, J., Send, U. & Wallace, D. W. R. Rapid transfer of oxygen to the deep ocean mediated by bubbles. *Nat. Geosci.* **13**, 232–237 (2020).
30. Koelling, J. et al. Progressive oxygenation of the North Atlantic subpolar gyre. *J. Geophys. Res. Oceans* **130**, e2024JC022157 (2025).
31. Talley, L. D. & McCartney, M. S. Distribution and circulation of Labrador Sea Water. *J. Phys. Oceanogr.* **12**, 1189–1205 (1982).
32. Wattenberg, H. Die durchlüftung des Atlantischen Ozeans: vorläufige mitteilung aus den ergebnissen der Deutschen Atlantischen Expedition. *J. du Conseil* **4**, 68–79 (1929).
33. Atamanchuk, D. et al. Linking oxygen and carbon uptake with the Meridional Overturning Circulation using a transport mooring array. *Oceanography* **34**, 9 (2021).
34. Miller, U. K. et al. Oxygen optodes on oceanographic moorings: recommendations for deployment and in situ calibration. *Front. Mar. Sci.* **11**, 1441976 (2024).
35. Fu, Y. et al. Characterizing the interannual variability of North Atlantic subpolar overturning. *Geophys. Res. Lett.* **52**, e2025GL114672 (2025).
36. Duyck, E., Foukal, N. P. & Frajka-Williams, E. Circulation of Baffin Bay and Hudson Bay waters on the Labrador shelf and into the subpolar North Atlantic. *Ocean Sci.* **21**, 241–260 (2025).
37. Tooth, O. J., Johnson, H. L. & Wilson, C. Controls on dense-water formation along the path of the North Atlantic subpolar gyre. *Ocean Sci.* **21**, 2101–2123 (2025).
38. Rhein, M. et al. Deep water formation, the subpolar gyre, and the Meridional Overturning Circulation in the subpolar North Atlantic. *Deep Sea Res. II* **58**, 1819–1832 (2011).
39. DeVries, T. & Weber, T. The export and fate of organic matter in the ocean: new constraints from combining satellite and oceanographic tracer observations. *Glob. Biogeochem. Cycles* **31**, 535–555 (2017).
40. Falck, E. & Gade, H. G. Net community production and oxygen fluxes in the Nordic Seas based on O_2 budget calculations. *Glob. Biogeochem. Cycles* **13**, 1117–1126 (1999).
41. Lauvset, S. K. et al. Continued warming, salinification and oxygenation of the Greenland Sea gyre. *Tellus* **70**, 1476434 (2018).
42. Possenti, L. et al. Norwegian Sea net community production estimated from O_2 and prototype CO_2 optode measurements on a seaglider. *Ocean Sci.* **17**, 593–614 (2021).
43. Boers, N. Observation-based early-warning signals for a collapse of the Atlantic Meridional Overturning Circulation. *Nat. Clim. Change* **11**, 680–688 (2021).
44. van Westen, R. M., Kliphuis, M. & Dijkstra, H. A. Physics-based early warning signal shows that AMOC is on tipping course. *Sci. Adv.* **10**, eadk1189 (2024).
45. Rossby, T., Palter, J. & Donohue, K. What can hydrography between the New England slope, Bermuda and Africa tell us about the strength of the AMOC over the last 90 years? *Geophys. Res. Lett.* **49**, e2022GL099173 (2022).
46. Baker, J. A. et al. Continued Atlantic overturning circulation even under climate extremes. *Nature* **638**, 987–994 (2025).
47. Bonan, D. B. et al. Observational constraints imply limited future Atlantic Meridional Overturning Circulation weakening. *Nat. Geosci.* **18**, 479–487 (2025).
48. Le Bras, I. A.-A. & Timmermans, M.-L. Can the marked Arctic Ocean freshwater content increases of the last two decades be explained within observational uncertainty? *J. Geophys. Res. Oceans* **130**, e2024JC021061 (2025).
49. Yashayaev, I. Intensification and shutdown of deep convection in the Labrador Sea were caused by changes in atmospheric and freshwater dynamics. *Commun. Earth Environ.* **5**, 156 (2024).
50. Wei, X. & Zhang, R. Weakening of the AMOC and strengthening of Labrador Sea deep convection in response to external freshwater forcing. *Nat. Commun.* **15**, 10357 (2024).
51. Garcia, H. E. et al. *World Ocean Atlas 2023, Volume 3: Dissolved Oxygen, Apparent Oxygen Utilization Dissolved Oxygen Saturation and 30-Year Climate Normal* NOAA Atlas NESDIS 91 (NOAA National Centers for Environmental Information, 2024); https://doi.org/10.25923/rb67-ns53
52. Reagan, J. R. et al. *World Ocean Atlas 2023* (NOAA National Centers for Environmental Information, 2023); https://www.ncei.noaa.gov/archive/accession/NCEI-WOA23

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026

[1]Graduate School of Oceanography, University of Rhode Island, Narragansett, RI, USA. [2]Marine Chemistry and Geochemistry Department, Woods Hole Oceanographic Institution, Woods Hole, MA, USA. [3]MIT-WHOI Joint Program, Woods Hole, MA, USA. [4]Department of Oceanography, Dalhousie University, Halifax, Nova Scotia, Canada. [5]Department of Earth and Environmental Sciences, Boston College, Chestnut Hill, MA, USA. [6]College of Marine Science, University of South Florida, Tampa, FL, USA. [7]GEOMAR Helmholtz Centre for Ocean Research, Kiel, Germany. [8]School of Oceanography, University of Washington, Seattle, WA, USA. [9]Cooperative Institute for Climate, Ocean, and Ecosystems Studies, University of Washington, Seattle, WA, USA. [10]Physical Oceanography Department, Woods Hole Oceanographic Institution, Woods Hole, MA, USA. [11]Present address: Department of Earth and Atmospheric Sciences, Cornell University, Ithaca, NY, USA. ✉e-mail: umiller@cornell.edu

Methods

Calibration of moored oxygen data

The GOHSNAP oxygen time series dataset was carefully calibrated against shipboard oxygen casts collected on the 2020 deployment and 2022 recovery cruises. The shipboard oxygen casts themselves were calibrated against Winkler-titrated bottle samples, the gold standard of oxygen measurements in the ocean[53]. Validation against oxygen casts external to the calibration process as well as to available Biogeochemical Argo (BGC-Argo) profiles in the vicinity of the moorings suggest our calibrated oxygen dataset is characterized by an error of 2.5–3 μmol kg^{-1}, or ~1%. For full details on the calibration and validation of the GOHSNAP dataset, see ref. 34.

Gridding of moored oxygen measurements to the OSNAP line

Daily averages of discrete dissolved oxygen time series (at the location of the 'x's in Fig. 1c) were gridded to the depths and longitudes of the Labrador Sea OSNAP grid using a random forest regressor (SciPy's RandomForestRegressor). The regressor was trained to predict dissolved oxygen concentration (μmol kg^{-1}) from practical salinity, potential temperature (°C), depth (m), longitude, calendar year, bathymetric depth (m), and both the sine and cosine transformation of day of year (DOY) in radians.

During training of the random forest, a random 20% of the dissolved oxygen time series data were reserved for testing. The remaining 80% were further split using a random 80–20 split into a training and validation dataset, where the validation data were used to tune random forest hyperparameters, such as tree depth. Following methods modified from ref. 54, a random forest with 600 trees and a minimum leaf size of 10 was trained on the training dataset using a variety of different 'feature' combinations and maximum tree depths to determine the best configuration. First, all possible combinations of the following features were used: practical salinity, potential temperature, potential density, water column depth, longitude, latitude, DOY[55], DOY in radians (calculated as 2 × DOY/365.25), the sine or cosine of DOY in radians[54], days since the year 2000, calendar year, North Atlantic Oscillation index[49], and General Bathymetry Chart of the Oceans (GEBCO) bathymetry[56] at a resolution of 1/240° in both latitude and longitude. As the number of features used was variable, the log base 2 of the number of features was used to make a decision split (max_features).

The optimal feature combination (comprising practical salinity, potential temperature, depth, longitude, calendar year, bathymetric depth, and the sine and cosine transformation of DOY in radians) was selected based on its minimization of the root mean square error (RMSE) of the validation dataset. In order to minimize overfitting of the random forest to the training dataset, the optimal maximum tree depth was determined from a plot of the validation RMSE versus tree depth curve, which decays approximately exponentially. Using a common mathematical heuristic, the maximum tree depth of 12 was selected visually from the 'elbow' of the curve, representing the point at which the RMSE is no longer substantially reduced by increasing the complexity of the random forest. All other hyperparameters were left as default. Overall, the random forest performed with a test error of 2.9 μmol kg^{-1}, which we define as the algorithm uncertainty (Supplementary Fig. 6).

Evaluating uncertainties and bias of the gridded oxygen product

Quantifying random forest uncertainties. Although the random forest performs well (~1% error), the training dataset was spatially limited to the discrete depths and locations of the moorings. We compare the random forest with two external datasets in order to: (1) assess the ability of the random forest to extrapolate outside its training data and produce a gridded product that spans the entire Labrador Sea; and (2) quantify the uncertainty associated with this gridding process.

The first set of external oxygen data comprises any BGC-Argo floats that profiled within ±3° latitude and ±1.5° longitude of an OSNAP grid position during the optode deployment period of June 2020–July 2022 (Supplementary Fig. 7). The latitude and longitude bounds used to collate float profiles match those used by OSNAP to incorporate Argo float data into their temperature and salinity grids. A total of 219 profiles from 10 floats were obtained. The major assets of the BGC-Argo dataset are that it contains oxygen measurements (1) from the central regions of the Labrador Sea, outside the boundary currents in which the random forest training data (moored oxygen time series) were collected; and (2) that span the 2-year deployment period. However, the BGC-Argo floats are limited to profiling depths of 0–2,000 m. BGC-Argo oxygen data are calibrated either by comparison against in-air measurements made by the float when it surfaces or comparison against a deep value take from the World Ocean Atlas climatology, depending on the technical specification of each float[57].

The second set of external oxygen data consists of 341 conductivity–temperature–depth (CTD) oxygen profiles collected in June 2020 and July 2022 during six research cruises on which the moored optodes were either deployed or recovered: cruises AR46[58], AR69-01[59], AR45[60], AR69-03[61], MSM94[62] and M184[63]. Only profiles collected within 50 km of the OSNAP line between mooring sites were considered for this evaluation (Supplementary Fig. 7). The CTD oxygen profiles were carefully calibrated using high-quality Winkler titrations[34] and, in contrast to BGC-Argo profiles, capture the full depth of the water column. However, they are limited to the months of June and July.

For most of these external datasets, the random forest performed with relatively low error, with RMSEs (~3–8 μmol kg^{-1}) comparable to the algorithm uncertainty (~3 μmol kg^{-1}). The larger RMSE values were associated with CTD casts that sampled waters with oxygen concentrations greater than ~350 μmol kg^{-1}, which is outside the oxygen distribution of the training data (Supplementary Fig. 6). To better understand these errors, which are the difference in 'true' oxygen (measured by the BGC-Argo floats and CTD profile) and 'predicted' (random forest) oxygen, we looked at the error in density space. Examination of error as a function of density (Supplementary Fig. 8) shows that it is water-mass dependent, systematically underestimating oxygen values in the lighter density classes in the boundary currents, but exhibiting a relatively low error (~4–5 μmol kg^{-1}) in the denser water classes like LSW (Supplementary Fig. 8). Accordingly, we quantify this error as the reciprocal function fit to the absolute error between the true and random-forest-predicted oxygen binned by potential density. As this function increases exponentially with decreasing density, the error was capped at 40 μmol kg^{-1} (~10% error), which was approximately the largest error measured in these lighter density classes. We define this error as the gridding uncertainty.

The total uncertainty (u) in the gridded oxygen product (equation (1)), is defined as the sum of the gridding uncertainty, which is a function of density and includes the algorithm uncertainty ($u([O_2])_{RF}$; where the subscript RF refers to the random forest gridding method; see Supplementary Fig. 8), and temporal variability ($u([O_2])^2_{SE}$), quantified as the standard error (SE) of daily gridded oxygen values.

$$u([O_2])_{total} = \sqrt{u([O_2])^2_{RF} + u([O_2])^2_{SE}} \quad (1)$$

Equation (1) is used to calculate the uncertainty for each grid point of the monthly oxygen grid product.

Bias in the DSOW layer. Accurate representation of zonal (east–west) gradients by the gridded oxygen product is key to properly quantifying oxygen export out of the Labrador Sea. Because our moored oxygen dataset lacked measurements in the DSOW[64] layer within the Labrador Sea ($\sigma_\theta > 27.88$ kg m^3) due to instrument failure, we compared the zonal gradient predicted by the random forest for this water mass with World Ocean Atlas climatology. While the climatology coinciding with our mooring deployment time period showed no zonal oxygen gradient within DSOW in the Labrador Sea, our gridded oxygen from the random

forest showed that the western portion of the Labrador Sea had oxygen concentrations on average ~10 µmol kg^{-1} higher than in the east in the DSOW density class. This increase would suggest that the DSOW layer gains oxygen as it passes through the Labrador Sea. As the DSOW layer is not reached by deep convection in the Labrador Sea, this is not a physically realistic gradient and instead is likely to represent a bias in the random forest due to limited DSOW observations in the training data. To address this bias, a correction term was defined as the mean difference between oxygen predicted by the random forest and that from World Ocean Atlas climatology in the DSOW water class. This bias correction term was calculated for the west, central and east portions of the Labrador Sea and added to the oxygen grid to correct for the erroneous zonal oxygen gradient.

Calculation of oxygen transport

Monthly oxygen transport across the Labrador Sea OSNAP line is calculated as the product of the monthly gridded oxygen and the monthly gridded OSNAP velocity. When producing their gridded velocity product, OSNAP adds a small correction factor to each grid cell such that the net transport across the whole OSNAP line is 0 Sv, with an assigned transport across OSNAP west of −1.6 Sv (based on long-term estimates of transport through the Davis Strait) and a compensating transport of 1.6 Sv across OSNAP east[13]. To isolate the oxygen transport terms of interest here (isopycnal and AMOC transport) from that carried by the throughflow, a volume transport of −1.6 Sv must be removed from OSNAP west prior to multiplication by the GOHSNAP oxygen grid for the purpose of calculating oxygen transport. Zeroing out the throughflow is standard practice in AMOC calculations[13,65]. Similarly to OSNAP, we add a small correction factor (~0.0006 m s^{-1}) to each velocity grid cell such that net volume flux across OSNAP west is 0 SV each month. This 'corrected' monthly gridded grid, along with the monthly gridded oxygen, is then used to calculate net oxygen transport (T) according to:

$$T = \int_{x_w}^{x_e}\int_{z_{min}}^{z_{max}} O_2\, v\, dz\, dx \tag{2}$$

where O_2 is oxygen in mol m^{-3}, v is velocity in m s^{-1}, dx is the horizontal distance between grid points in metres, dz is the thickness of each grid layer in metres, x_w and x_e are the western and eastern boundaries of the Labrador Sea OSNAP line, respectively, and z_{min} and z_{max} are the shallowest and deepest grid depths in metres, respectively. Volume flux (v dx dz) is obtained from OSNAP.

To calculate the isopycnal and overturning components of the oxygen transports, the oxygen and velocity grids are first re-gridded in potential density coordinates at an interval of 0.01 kg m^{-3}, the same binning resolution used by OSNAP. Velocity in m s^{-1} and oxygen in mol m^{-3} are first decomposed into a zonal mean along each isopycnal layer, denoted by angled brackets, and the perturbations from this mean, denoted by primes (for example, refs. 13,28,65).

$$v = \langle v\rangle(n,t) + v'(x,n,t) \tag{3}$$

$$O_2 = \langle O_2\rangle(n,t) + O_2'(x,n,t) \tag{4}$$

where x is zonal distance along the OSNAP line, n is the isopycnal layer and t is time. The overturning and isopycnal components of oxygen transport are then defined as

$$T_{overturning} = \int_{z_{\sigma_{min}}}^{z_{\sigma_{max}}} \langle O_2\rangle\langle v dx\rangle\, dz \tag{5}$$

$$T_{isopycnal} = \int_{x_w}^{x_e}\int_{z_{\sigma_{min}}}^{z_{\sigma_{max}}} O_2'\, v'\, dz\, dx \tag{6}$$

where x_w and x_e are the western and eastern boundaries of the Labrador Sea OSNAP line, respetively, dx is the horizontal distance between grid points in metres, dz is the thickness of each isopycnal layer in metres, and σ_{min} and σ_{max} are the minimum and maximum potential density on the section.

Estimate of uncertainty in oxygen transport

Uncertainty in the oxygen transport grid (equation (2)) is estimated using a Monte Carlo approach (see Code availability). For each iteration of the oxygen transport calculation, realizations of the oxygen concentration and velocity grids are chosen randomly from their respective normal distributions. For monthly oxygen, the normal distribution is built from the monthly average and error for each grid point, with the latter estimated according to equation (1). The same is done for monthly velocity, with the standard error (SE) for each grid point calculated as

$$SE = \frac{\sigma}{\sqrt{30\ \text{days}/(2\tau)}} \tag{7}$$

where σ is standard deviation of daily velocities calculated over each month and τ is the integral timescale in days, calculated from the autocorrelation time function of the time series at each grid point, as in ref. 13. Thirty days is the average number of data points contained in a monthly average and the division by 2τ accounts for the fact that time series are often autocorrelated, thus reducing the number of independent data points in a sample. We acknowledge that there are possible sources of uncertainty[49] in the OSNAP velocity field not captured by equation (7). Monte Carlo iterations were performed until the running average of calculated monthly oxygen transport converged to that calculated by equation (2). Uncertainty in the calculated oxygen transport was then defined as the monthly standard deviation calculated across all Monte Carlo iterations.

Respiration estimates

In order to provide an estimate of respiration rates within the LSW layer, we utilized simulated biological respiration rates from SIMPLE-TRIM[39], a data-assimilating biogeochemical model embedded in a steady-state ocean circulation inverse model. Because the available model output does not include density or salinity, we used a temperature range of 3–4 °C to define the LSW water mass[66]. This modelled water mass defined using temperature largely follows the known circulation pathways of LSW (Supplementary Fig. 4). A temperature range of 3–6 °C was chosen to approximate respiration across LSW and the overlying low-oxygen thermocline waters with which it mixes (Supplementary Fig. 5). Mixing with underlying, similarly high-oxygen overflow waters would not contribute substantially to oxygen loss from the LSW layer and is therefore excluded from this exercise. The surface layer (approximately 0–200 m depth) is also excluded as oxygen here is sourced from photosynthesis and air–sea interaction, in addition to outcropping LSW. To estimate total respiration across the North Atlantic, respiration rates were summed across depth and a geographical range bounded by 0–50° N and 280–340° E (see Code availability). For the eastern subpolar North Atlantic, this geographical range was defined as 50–80° N and east of 306° E.

Data availability

The calibrated optode time series data and gridded oxygen field are available via the Biological and Chemical Oceanography Data Management Office (BCO-DMO) data repository at https://doi.org/10.26008/1912/bco-dmo.986667.1 (ref. 67) and https://doi.org/10.26008/1912/bco-dmo.995975.1 (ref. 68), respectively. The 2020–2022 OSNAP gridded monthly velocity, temperature and salinity fields are available via the Georgia Tech repository at https://doi.org/10.35090/gatech/78023 (ref. 69). SIMPLE-TRIM ecosystem model output is freely obtained from https://tdevries.eri.ucsb.edu/models-and-data-products/.

Code availability

Code for calculating oxygen transport and net respiration estimates is available via Zenodo at https://doi.org/10.5281/zenodo.19432748 (ref. 70) and for producing the gridded oxygen field at https://doi.org/10.5281/zenodo.16577485 (ref. 71).

References

53. Langdon, C. in *The GO-SHIP Repeat Hydrography Manual: A Collection of Expert Reports and Guidelines. Version 1* (eds Hood, E. M. et al) IOCCP Report Number 14; ICPO Publication Series Number 134 (GO-SHIP, 2010); https://doi.org/10.25607/OBP-1350
54. Sharp, J. D. et al. GOBAI-O_2: temporally and spatially resolved fields of ocean interior dissolved oxygen over nearly 2 decades. *Earth Syst. Sci. Data* **15**, 4481–4518 (2023).
55. Sauzéde, R. et al. A neural network-based method for merging ocean color and Argo data to extend surface bio-optical properties to depth: retrieval of the particulate backscattering coefficient. *J. Geophys. Res. Oceans* **121**, 2552–2571 (2016).
56. Schenke, H. W. in *Encyclopedia of Marine Geosciences* (eds Harff, J., Meschede, M., Petersen, S. & Thiede, J.) 268–269 (Springer, 2016); https://doi.org/10.1007/978-94-007-6238-1_63
57. Thierry, V. & Bittig, H. C. *Argo Quality Control Manual for Dissolved Oxygen Concentration* (Argo-BGC group, 2021); https://archimer.ifremer.fr/doc/00354/46542/
58. Bigorre, S. & Furey, H. H. Cruise AR46 on RV Neil Armstrong. *R2R* version number 1 https://doi.org/10.7284/908990 (2021).
59. White, S. Cruise AR69-01 on RV Neil Armstrong. *R2R* version number 1 https://doi.org/10.7284/909943 (2023).
60. Pickart, R. Cruise AR45 on RV Neil Armstrong. *R2R* version number 1 https://doi.org/10.7284/908988 (2021).
61. Straneo, F. Cruise AR69-03 on RV Neil Armstrong. *R2R* version number 1 https://doi.org/10.7284/909945 (2023).
62. Karstensen, J. et al. *Western Subpolar North Atlantic Transport Variability, Cruise No. MSM94, 02. August - 06. September 2020, Emden (Germany) - Emden (Germany)* (Gutachterpanel Forschungsschiffe, 2020).
63. Karstensen, J. et al. *Understanding Processes and Long-term Variability in the Labrador Sea, Cruise No. M184, 12. August 2022 - 15. September 2022, St. John's (Canada) - Hamburg (Germany), LabSeaVar 2022* (Begutachterpanel Forschungsschiffe, 2023); https://oceanrep.geomar.de/id/eprint/60759/
64. Tanhua, T., Olsson, K. A. & Jeansson, E. Formation of Denmark Strait Overflow Water and its hydro-chemical composition. *J. Mar. Syst.* **57**, 264–288 (2005).
65. Holliday, N. P. et al. Subpolar North Atlantic overturning and gyre-scale circulation in the summers of 2014 and 2016. *J. Geophys. Res. Oceans* **123**, 4538–4559 (2018).
66. Wright, W. R. & Worthington, L. V. in *Serial Atlas of the Marine Environment* Folio 19 (American Geographical Society, 1970).
67. Miller, U. K. et al. Dissolved oxygen time series measured on the Overturning in the Subpolar North Atlantic Program (OSNAP) moorings within the boundary currents of the Labrador and Irminger Seas from summer 2020 to summer 2022. Version 1. *BCO-DMO* https://doi.org/10.26008/1912/bco-dmo.986667.1 (2025).
68. Park, E., Nicholson, D. P., Miller, U. K. & Palter, J. B. A monthly gridded oxygen product for a Labrador Sea basin transect from August 2020 to July 2022. *BCO-DMO* https://doi.org/10.26008/1912/bco-dmo.995975.1 (2026).
69. Fu, Y. et al. Meridional Overturning Circulation observed by the Overturning in the Subpolar North Atlantic Program (OSNAP) array from August 2014 to July 2022. *Georgia Tech Library* https://doi.org/10.35090/gatech/78023 (2025).
70. Miller, U. K. unamiller/GOHSNAP-oxygen-transport: v2. *Zenodo* https://doi.org/10.5281/zenodo.19432748 (2026).
71. Park, E., Nicholson, D., Miller, U. K. & Palter, J. Trained random forest for GOHSNAP oxygen grid. *Zenodo* https://doi.org/10.5281/zenodo.16577485 (2025).

Acknowledgements

We thank the Overturning in the Subpolar North Atlantic Program (OSNAP), and in particular D. Hebert, M. Ringuette, S. Lozier, F. Straneo, A. Bower and R. Pickart for supporting us in deploying optodes on the mooring array. We are grateful to the crew and scientists aboard the 2020 and 2022 OSNAP mooring cruises (AM2001, AR45, AR46, MSM94-01, AT48-05, AR69-03 and M184), and to N. Renier for Fig. 2.

Author contributions

J.P. conceptualized and administered the project. Together with D.A., I.L.B., D.N. and H.P., J.P. acquired funding for and supervised the project. U.K.M., J.P., E.P. and D.N. developed the methodology. U.K.M. and E.P. led software development and visualization and, together with H.P. and K.F., curated data. U.K.M. wrote the original paper draft and U.K.M., J.P., E.P., D.A., K.F., Y.F., J. Karstensen, J. Koelling, I.L.B., H.N., D.N., H.P. and M.Y., reviewed and edited the manuscript.

Funding

U.K.M. and J.P. were supported by National Science Foundation (NSF) awards 1947829 and J.P. also by NSF asward 2446499. H.P., K.F. and M.Y. were supported by NSF award 1947970. I.L.B. and H.N. were supported by NSF awards 2038481 and 2122579. D.A. and J. Koelling were supported by the Canada Excellence Research Chair (CERC) in Ocean Science and Technology, the Canada First Research Excellence Fund (Ocean Frontier Institute grant, Module B) and the CERC. Ocean group led by D. Wallace at Dalhousie University. J. Koelling was additionally supported by the Cooperative Institute for Climate, Ocean, & Ecosystem Studies (CICOES) under NOAA Cooperative Agreement NA20OAR4320271, Contribution No. 2025-1467. E.P. was supported by NSF award 2023080. E.P. and D.N. were supported by award NSF 1947567. OSNAP data were collected and made freely available by the OSNAP project and all the national programmes that contribute to it (www.o-snap.org). The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41561-026-02057-3.

Correspondence and requests for materials should be addressed to Una Kim Miller.

Peer review information *Nature Geoscience* thanks the anonymous reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available. Primary Handling Editor: James Super, in collaboration with the *Nature Geoscience* team.

Reprints and permissions information is available at www.nature.com/reprints.